ModelIt!

Teacher's Guide

Can We Engineer the Climate?

Modeling Geoengineering Technologies, Carbon Capture, and Ethical Dimensions

Quick Reference

Grade Level	Grade 12L2
NGSS Standard	HS-ESS2-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ESS2-4

Use a model to describe how variations in the flow of energy into and out of Earth\'s systems result in changes in climate.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop models of Earth\'s energy balance to predict how geoengineering interventions would alter climate system dynamics, including unintended consequences like regional disruption and termination shock.
Disciplinary Core Idea	**ESS2.D Weather and Climate** — The greenhouse effect and energy balance of Earth are determined by atmospheric composition and solar input. Geoengineering proposals alter either the input (SRM) or the composition (CDR), with fundamentally different implications for climate system stability.
Crosscutting Concept	**Cause and Effect** — Students analyze the causal relationships between geoengineering interventions and their climate system effects, distinguishing between addressing root causes (reducing CO2) and masking symptoms (reflecting sunlight), and identifying the unintended cascading effects of each approach.

CAST Testing Alignment

MULTIPLE CHOICE:

A nation deploys stratospheric aerosol injection for 30 years, successfully keeping Global Mean Temperature stable at 1.5 degrees above pre-industrial. During this time, atmospheric CO2 has risen from 424 to 550 ppm. What would happen if the aerosol injection were suddenly stopped? A) Temperature would remain stable because the climate has adapted B) Temperature would gradually increase over 50 years to the level expected at 550 ppm C) Temperature would rapidly increase by several degrees within a decade as the full greenhouse effect of 550 ppm is unmasked D) The ocean would absorb the excess CO2 and prevent warming

CONSTRUCTED RESPONSE:

Using your climate intervention model, compare the long-term climate outcomes of two strategies: (1) deploying SRM while continuing current emissions, and (2) reducing emissions by 50% while deploying 5 Gt/year of CDR. Reference at least four model components including Ocean Acidification, Termination Risk, and Carbon Removal Rate in your analysis of why these strategies produce fundamentally different outcomes.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

As the gap between climate targets and actual emission reductions widens, attention is increasingly turning to geoengineering — deliberate large-scale interventions in Earth\'s climate system. These proposals range from reflecting sunlight back to space to pulling CO2 directly from the atmosphere. The technologies are fascinating, controversial, and fraught with risks and ethical dilemmas that go far beyond the science. Understanding geoengineering is essential for every scientifically literate citizen because these decisions may define the 21st century.

Solar Radiation Management: The Sunshade Approach

SRM techniques aim to reduce the amount of sunlight reaching Earth\'s surface. The most discussed approach is stratospheric aerosol injection (SAI) — releasing reflective sulfate particles into the upper atmosphere that scatter incoming sunlight, mimicking the cooling effect of large volcanic eruptions. Mount Pinatubo\'s 1991 eruption injected 20 million tons of sulfur dioxide into the stratosphere and cooled the planet by approximately 0.5 degrees Celsius for two years. SAI would replicate this effect continuously using a fleet of high-altitude aircraft. Cost estimates are surprisingly low — $2-10 billion per year to offset 1-2 degrees of warming. But SRM does not remove CO2: it masks warming while the root cause continues worsening, like taking painkillers without treating an infection.

Carbon Dioxide Removal: Addressing the Root Cause

CDR technologies extract CO2 from the atmosphere and store it permanently. Direct Air Capture (DAC) uses chemical processes to bind CO2 from ambient air, then compresses and injects it underground into geological formations. The technology works — Climeworks in Iceland operates the world\'s largest DAC plant, removing 36,000 tons of CO2 per year. But the scale challenge is staggering: annual emissions are 36 billion tons, meaning current DAC removes 0.001% of annual emissions. Scaling to 10 billion tons per year would require thousands of facilities, enormous energy input (potentially dedicated renewable farms), and trillions of dollars in investment. Enhanced weathering — accelerating the natural process by which minerals absorb CO2 — and ocean alkalinity enhancement offer potentially cheaper paths but face ecological uncertainty. CDR addresses the fundamental problem but cannot operate at meaningful scale for decades.

The Termination Problem: A Multigenerational Trap

The most dangerous aspect of SRM is what happens when it stops. While SRM is deployed, CO2 continues accumulating in the atmosphere. If SRM masks 2 degrees of warming for 50 years, stopping it would expose the planet to the full warming effect of all the CO2 that accumulated behind the solar shield — a rapid temperature increase of several degrees within a decade, far faster than any natural climate change. This rate of change would be catastrophic for ecosystems and agriculture. Termination shock means SRM creates a commitment that must be maintained for centuries — until CO2 levels are naturally or artificially reduced to safe levels. But maintaining a global technological system for centuries requires institutional stability that no human civilization has ever demonstrated. Wars, economic crises, pandemics, or political changes could interrupt deployment with devastating consequences.

Governance and Justice: Who Decides for the Planet?

Perhaps the most challenging dimension of geoengineering is governance. SRM affects the entire planet but would be deployed by a few nations — potentially just one. Regional effects would not be evenly distributed: models suggest SAI could reduce monsoon rainfall in South and East Asia, affecting the food production of billions. Nations that benefit from deployment may differ from those that suffer its side effects, creating a new dimension of climate inequity. Currently, no international treaty governs geoengineering, and no consensus exists on whether research, testing, or deployment should proceed. The moral hazard concern adds urgency: if geoengineering makes climate change seem less threatening, governments may reduce efforts to cut emissions, ultimately making the underlying problem worse while adding the new risks of intervention.

4 The LEVER Framework

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify Atmospheric CO2, Solar Radiation Reaching Surface, Global Mean Temperature, Ocean Acidification Level, Regional Climate Disruption, Carbon Removal Rate, and Termination Risk as key components of the climate intervention system.

E **Establish Relationships**

Students determine that SRM reduces temperature without addressing CO2 or acidification, that CDR addresses the root cause but at inadequate scale, that SRM creates growing termination risk over time, and that only emission reduction combined with CDR achieves genuine climate stabilization.

V **Visualize & Model**

Students build a computational model comparing how SRM and CDR differently affect Earth\'s energy balance, ocean chemistry, and long-term climate stability, revealing the fundamental asymmetry between masking symptoms and treating causes.

E **Evaluate Outcomes**

Students run SRM deployment, CDR scale-up, and combined emission reduction with CDR scenarios to test predictions about effectiveness, risks, and the conditions for genuine climate stabilization.

R **Revise & Extend**

Students add research governance gaps, ozone layer interactions, or intergenerational equity to explore how additional dimensions affect the risk-benefit calculus and governance requirements of geoengineering.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Can we engineer the climate? The technology exists. The question is: should we? Today we model the science and confront the ethics."*

Watch For

Show cover image. Quick poll: If you could push a button that cooled the planet but didn\'t remove any CO2, would you push it? Take responses and reasoning — revisit after modeling.

Slide 2: Learning Objectives

3 min

Say: *"Today is as much about ethics and governance as it is about science. We\'re modeling technologies that could save the planet — or create new disasters."*

Watch For

Have students read objectives. Pre-teach vocabulary. Key distinction: SRM masks symptoms without treating the disease. CDR treats the disease but at tiny scale. This distinction will drive everything.

Slide 3: Big Question

3 min

Say: *"Who gets to decide to modify the climate for 8 billion people? What if the intervention helps some nations but harms others? Write your initial position."*

Watch For

Quick-write: Students write whether they think geoengineering should be pursued and why. Save for comparison — many positions will shift after modeling the termination problem.

Slide 4: LEVER Framework

2 min

Say: *"We\'ll build a model that compares two fundamentally different approaches to the climate crisis. One treats the symptom. One treats the cause. The model will reveal why this difference matters enormously."*

Watch For

Preview LEVER steps. Emphasize: this model will reveal a trap — a situation where the seeming solution creates a new and potentially worse problem.

Slide 5: Activity 1: Components

8 min

Say: *"Which components are external inputs we can control, and which are system responses? Think about what we can manipulate versus what the Earth system does in response."*

Watch For

Guide sorting: External = Atmospheric CO2 (emissions + CDR), Solar Radiation Reaching Surface (SRM can modify). Internal = the five climate system responses. Discuss: Why is Termination Risk internal? It\'s a consequence of deployment decisions, not a controllable input.

Slide 6: Activity 2: Connections

8 min

Say: *"Draw two pathways: SRM reduces solar radiation but doesn\'t touch CO2 — what happens to Ocean Acidification and Termination Risk? CDR reduces CO2 directly — how does this differ?"*

Watch For

Students draw arrows for both intervention types. Key discovery: SRM creates a dangerous divergence — temperature goes down but CO2, acidification, and termination risk go up. CDR aligns everything in the right direction but at tiny scale. Ask: Which is actually safer?

Slide 7: Activity 3: Simulation

10 min

Say: *"Run all three scenarios. The SRM scenario may look good at first — but watch what happens to Ocean Acidification and Termination Risk as the years pass. This is the trap."*

Watch For

Students run scenarios. Key discussion: In the SRM scenario, temperature stabilizes — but everything else gets worse. After 50 years, termination would cause catastrophic rapid warming. Ask: Is this a solution or a different kind of catastrophe?

Slide 8: Discoveries

5 min

Say: *"Our model reveals a fundamental truth: there is no shortcut past emission reduction. SRM is a trap that makes the underlying problem worse. CDR is real but tiny. The only genuine solution is reducing emissions."*

Watch For

Class discussion. Revisit initial poll: Has anyone changed their position on whether to push the button? Why? The termination problem and ocean acidification should have changed perspectives.

Slide 9: STEM Challenge

12 min

Say: *"The UN has asked your team to design the framework for deciding whether geoengineering should be deployed. This is about governance, justice, and science together. Who decides, and how?"*

Watch For

Groups design governance frameworks incorporating scientific risk criteria, consent mechanisms, equity principles, and termination protocols. Present and debate frameworks, addressing challenges from other groups.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Atmospheric CO2 Concentration, Solar Radiation Reaching Surface
- **INTERNAL:** Global Mean Temperature, Ocean Acidification Level, Regional Climate Disruption, Carbon Removal Rate, Termination Risk

Relationships

- **Atmospheric CO2 Concentration to Global Mean Temperature:** POSITIVE (+) — Higher atmospheric CO2 strengthens the greenhouse effect, trapping more outgoing radiation and increasing global temperature. This is the fundamental mechanism of climate change that CDR addresses directly.
- **Solar Radiation Reaching Surface to Global Mean Temperature:** POSITIVE (+) — More solar radiation reaching the surface increases energy input and temperature. SRM reduces this input, counteracting greenhouse warming — but only the temperature effect, not the chemical effects of elevated CO2.
- **Atmospheric CO2 Concentration to Ocean Acidification Level:** POSITIVE (+) — Higher atmospheric CO2 increases the amount dissolved in oceans, forming carbonic acid and reducing pH. SRM does not affect this pathway because it does not reduce CO2 — acidification continues regardless of SRM deployment.
- **Solar Radiation Reaching Surface to Regional Climate Disruption:** POSITIVE (+) — Reducing solar input unevenly across the planet — as SRM does — alters temperature gradients that drive weather patterns, potentially disrupting monsoons, shifting precipitation belts, and changing storm tracks.
- **Atmospheric CO2 Concentration to Termination Risk:** POSITIVE (+) — Higher CO2 behind the SRM shield means greater warming exposure if SRM stops. Termination Risk grows every year that SRM masks accumulating CO2, creating an increasingly dangerous dependency.

Key Discoveries

- Solar Radiation Management masks warming without addressing the root cause — the model shows that while SRM can stabilize temperature, Ocean Acidification continues worsening because atmospheric CO2 is unchanged, and Termination Risk grows every year of deployment
- Current Carbon Dioxide Removal capacity (0.01 Gt/year) is approximately 3,600 times smaller than annual emissions (36 Gt/year) — the model reveals the enormous gap between CDR aspiration and reality that makes it a complement to emission reduction, not a substitute
- The termination shock problem creates a multigenerational trap — if SRM is deployed for decades while CO2 continues accumulating, stopping it would cause several degrees of warming in under a decade, an unprecedented rate of change that would devastate ecosystems and human systems
- The most effective and lowest-risk pathway combines aggressive emission reduction with moderate CDR deployment — this approach addresses the root cause (CO2) rather than masking symptoms (temperature), avoids termination risk, and does not create regional climate disruption

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Geoengineering can solve climate change without reducing emissions"	Both SRM and CDR have fundamental limitations that make them supplements to, not substitutes for, emission reduction. SRM masks warming without addressing CO2, meaning ocean acidification continues and termination risk grows indefinitely. CDR operates at a scale roughly 3,600 times too small and would take decades and trillions of dollars to scale meaningfully. The model clearly shows that the only pathway to genuine climate stabilization requires aggressive emission reduction as the primary strategy, with CDR as a complement. Geoengineering without decarbonization is like pumping water out of a sinking boat without plugging the hole.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Solar radiation management is safe because volcanic eruptions have the same effect naturally"	While SRM mimics the cooling effect of volcanic eruptions, there are critical differences. Volcanic cooling is temporary (1-2 years) and uncontrolled; SRM would need to be continuous and sustained for centuries. Volcanic eruptions do cause regional climate disruption — Pinatubo\'s eruption significantly reduced monsoon rainfall and caused crop failures in some regions. Sustaining that effect indefinitely would compound regional impacts year after year rather than allowing recovery. And volcanic eruptions do not create termination risk because the cooling fades naturally; SRM termination would expose decades of accumulated warming instantly.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Carbon capture is too expensive and will never work at scale"	While current CDR costs are high ($250-600 per ton of CO2 for direct air capture), costs are declining rapidly — similar to the trajectory of solar panels, which fell 99.7% from their initial cost. Enhanced weathering and ocean alkalinity enhancement offer potentially cheaper pathways. The more accurate critique is that CDR at the scale needed (10+ Gt/year) requires enormous investment and decades to build — making it a long-term tool rather than an immediate solution. This does not mean CDR will never work; it means we cannot wait for it and must reduce emissions now while building CDR capacity for the future.